

Taichi Kamei

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EDUCATION

3rd Year Engineering Physics

Expected Graduation - May 2028

University of British Columbia | Vancouver

Relevant Courses: Solid and Fluid Mechanics, Mechanical Designs, Electric Circuit Analysis, Signals and Systems, Thermodynamics, Statistical Mechanics, Quantum Mechanics

EXPERIENCE

Firmware & Validation Engineer Co-op

May 2026 - Present

UBC Blusson Quantum Matter Institute

- Developing full-stack C++ firmware for 4-channel low-noise SNSPD current source, achieving lower noise spectral density than commercial SMU
- Implementing 24-bit ADC and 16-bit DAC SPI driver on ESP32 using ESP-IDF, enabling precision current measurement and control
- Designing FSM with FreeRTOS dual-core task architecture and ISR-driven sampling
- Developing automated V-I sweep to characterize SNSPD switching current and determine optimal bias point
- Building Python Qt6 GUI and CSV data logging for instrument and data acquisition
- Testing and verifying the noise floor, negative rail voltage noise, time domain voltage drift characteristics, and DAC&ADC accuracy of the instrument using oscilloscope and pico-ammeter
- Designing a current source PCB casing on SolidWorks, reducing vibrational noise coupling into the PCB

PROJECTS

Drone Flight Controller

Feb 2026 - Present

- Designed a power board for 4S LiPo, integrating BMS IC and 5V/5A buck converter
- Designed a custom 4-layer GPS & magnetometer PCB on KiCAD fitting within 260mm by 260mm
- Prototyping a flight controller PCB, integrating ESP32-S3-Mini-1U, BMS, IMU, barometer, and a RF transceiver using SX1261

Autonomous Clue Detecting Robot

November 2025

- Developed an autonomous vehicle in Gazebo and ROS 1 using OpenCV for PID control driving, obstruction avoidance, and CNN clue detection
- Designed and implemented a finite state machine (FSM) capable of transitioning its states for moving obstruction detection, clue board detection, and drive under various track surface conditions
- Optimized a PID algorithm for precise center-lane driving using dual side lines instead of a typical single line-following, overcoming camera center offset from the geometric center of the lane
- Created a Qt5 controller GUI which integrates simulation controls and launch of two ROS node scripts, boosting team productivity by centralizing all controls into a single window which normally takes 4+ separate terminals

Autonomous Payload Retrieving Robot

June - August 2025

- Designed a 3-DOF four-bar-linkage arm and arm base on Onshape and fabricated components using 3D printer, water jet cutter, and laser cutter
- Sized appropriate servo motors for the arm from maximum payload and bending moment calculations
- Implemented magnetic encoder-based rotational control for the arm base, achieving accuracy of less than 1° error
- Designed a custom 2-layer I2C multiplexer and I2C buffer PCB on KiCAD to solve peripheral address conflicts and signal degradation
- Prototyped a payload detection algorithm in Python on Raspberry Pi using two 2D LiDAR, generating depth and reflectance map in 7Hz. Implemented it in C++ on ESP32 with 15 Hz real-time detection

SKILLS

Software	Python, Java, C/C++, Assembly, VHDL, Linux, Git
Embedded	ESP-IDF, FreeRTOS, Arduino, Raspberry Pi, FPGA, ROS, I2C, SPI, UART
Electrical	Kicad, Altium, LTSpice, Soldering, Oscilloscope, DMM, Spectrum Analyzer, Logic Analyzer
Mechanical	Onshape, Siemens NX, 3D printing, Laser Cutting, Water Jet Cutting, Drill Press, Caliper